



DESCRIPTION

The PROTO216 A9 Template Board is a demonstration board with the recommended PROTO216 pinout for the development of new A9-size boards. You can use this template to write all technical documentation for a new PROTO216 board.

Like any PROTO216 PCB, this A9-size board can be fitted in any other breadboard, as it has the minimum standard size to fit in a socket. Manufacturing data and engineering documentation can be found in the [PROTO216 repository](#), or at the official [PROTO216 website](#).

Table 1. Electrical characteristics

SYMBOL	PARAMETER	NOTES	MIN	TYP	MAX	UNITS
V _{in}	Input voltage	V1 / V2	5	12	30	V
V _{out}	Output voltage	J1AS pins	0	-	5	V
I _{in}	Input current	V1 / V2	-	-	5	A
I _{out}	Output current	J1AS pins	-	-	1	A

BOARD IMAGE

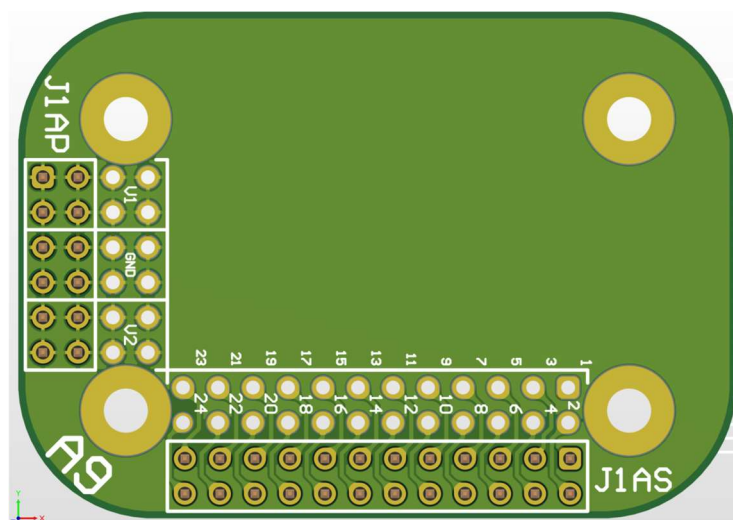


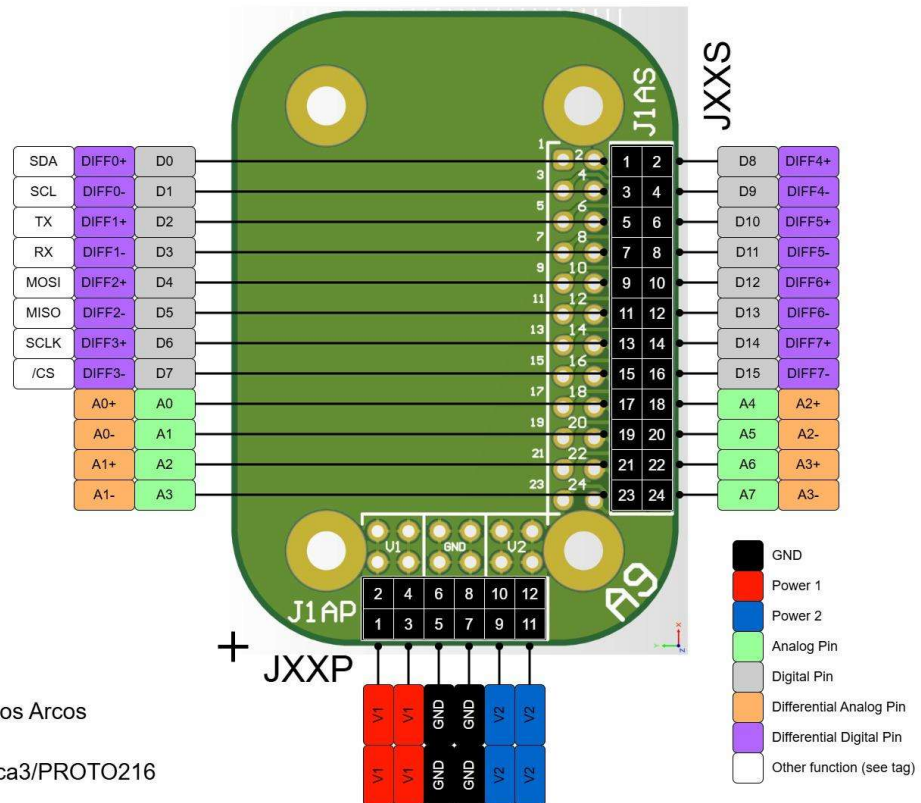
Figure 1. A9 TEMPLATE Board

BOARD PINOUT

The A9 Template Board has two connectors, called J1AP and J1AS. Following the PROTO216 standard pinout, J1AP groups power pins and J1AS groups signal pins.

PROTO216 PINOUT DIAGRAM

PROTO216 can be configured with any pin distribution. This diagram shows the recommended distribution to increase compatibility with other commercial solutions like Arduino or Raspberry Pi.



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Figure 2. A9 TEMPLATE Pinout

Table 2. J1AP Pinout

Pin number	Pin description	Pin number	Pin description
1	Voltage input 1 (V1)	7	Ground (GND)
2	Voltage input 1 (V1)	8	Ground (GND)
3	Voltage input 1 (V1)	9	Voltage input 2 (V2)
4	Voltage input 1 (V1)	10	Voltage input 2 (V2)
5	Ground (GND)	11	Voltage input 2 (V2)
6	Ground (GND)	12	Voltage input 2 (V2)

Table 3. J1AS Pinout

Pin number	Pin description	Pin number	Pin description
1	SDA / DIFF0+ / D0	13	SCLK / DIFF3+ / D6
2	DIFF4+ / D8	14	DIFF7+ / D14
3	SCL / DIFF0- / D1	15	/CS / DIFF3- / D7
4	DIFF4- / D9	16	DIFF7- / D15
5	TX / DIFF1+ / D2	17	A0+ / A0
6	DIFF5+ / D10	18	A2+ / A4
7	RX / DIFF1- / D3	19	A0- / A1
8	DIFF5- / D11	20	A2- / A5
9	MOSI / DIFF2+ / D4	21	A1+ / A2
10	DIFF6+ / D12	22	A3+ / A6
11	MISO / DIFF2- / D5	23	A1- / A3
12	DIFF6- / D13	24	A3- / A7

CONFIGURATION SETTINGS

This is a template board, and therefore no software or firmware can be set up here. If your board implements any kind of microcontroller or configurable logic, add it here.

It also applies to hardware parameters, as jumpers, switches, pins...

SCHEMATIC DIAGRAM

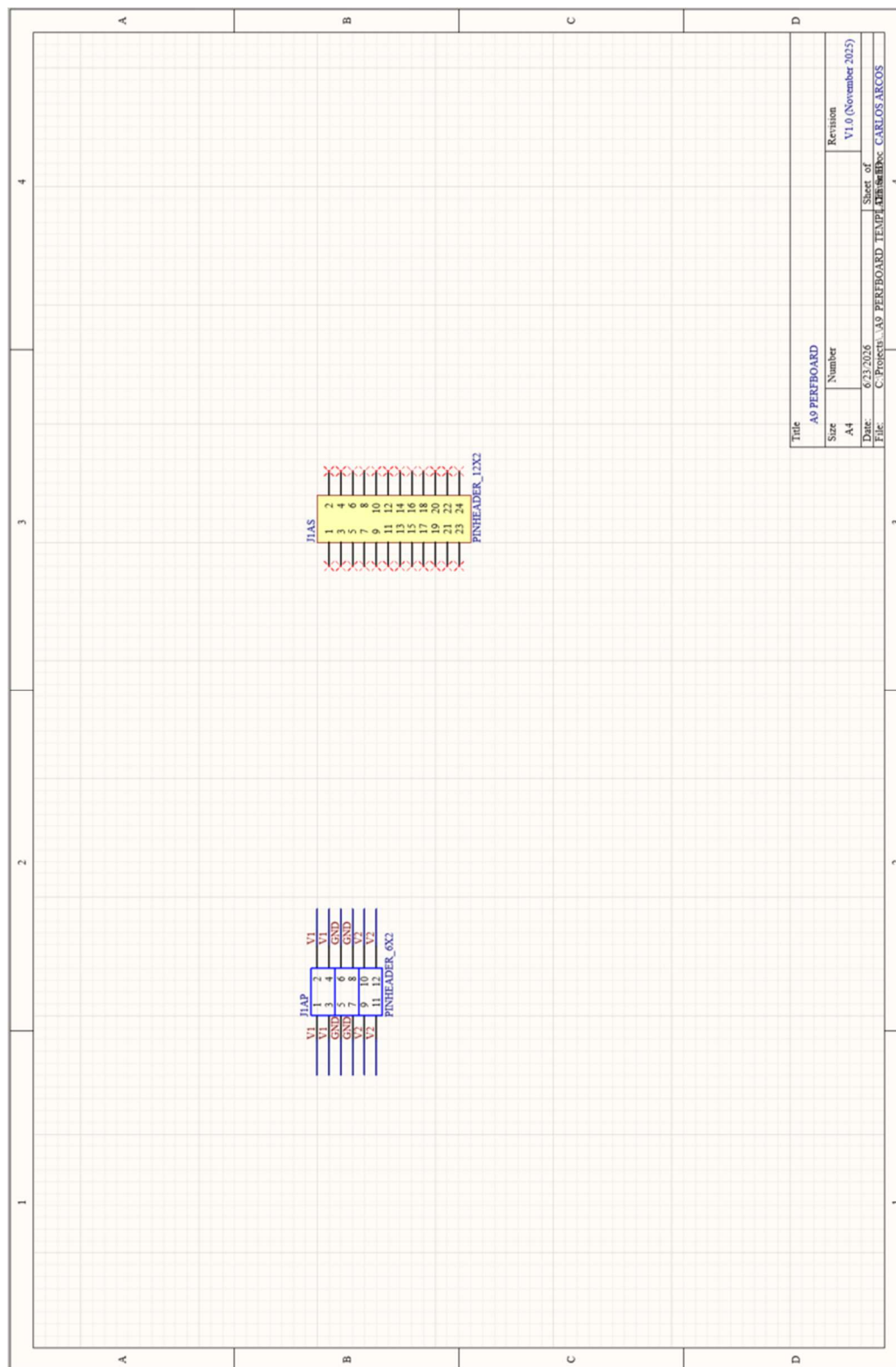


Figure 3. A9 TEMPLATE Schematic

MECHANICAL DIMENSIONS

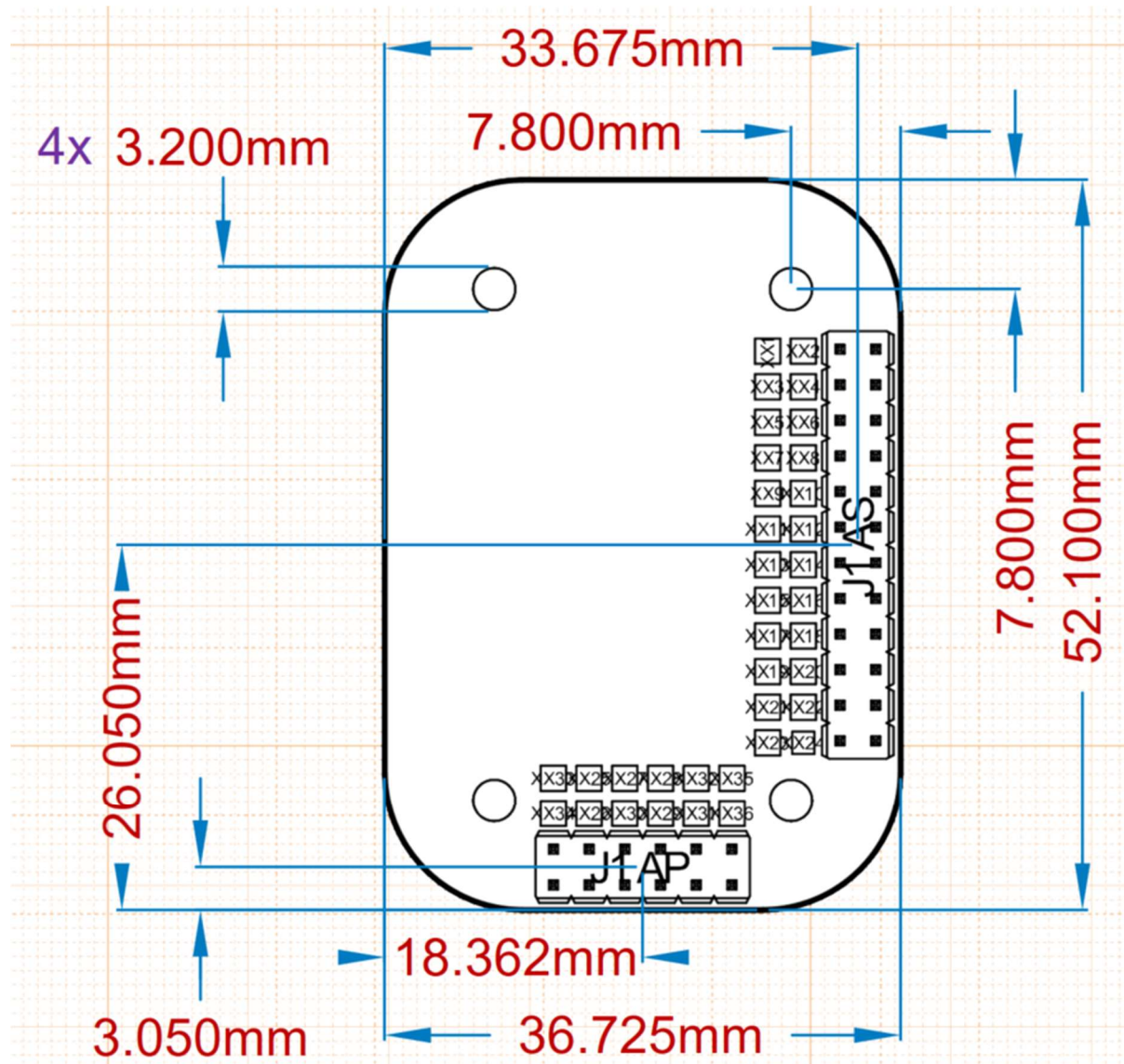


Figure 4. A9 TEMPLATE mechanical dimensions

LICENSE

This project is open-source hardware. We believe in the power of collaboration and the importance of keeping modular design standards accessible to everyone.

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We encourage you to experiment, iterate, and share your own modules. Whether you are improving an existing board or creating a new form-factor, your contributions help grow the PROTO216 community.

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